

Automatic Detection of Leukemia through Convolutional Neural Network

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Abstract—Leukemia is a fatal cancer disease that develops in blood-forming tissue by the excessive development of white blood cells (WBCs) in the human body. However, a bone marrow test is recommended by the pathologist to diagnose leukemia and further types of leukemia. Leukemia has two classes i.e., acute and chronic leukemia. Therefore, early leukemias detection enables preventative actions to be taken to avoid any harm to human life. In addition, several manual and automatic methods have been proposed, however, they possess some drawbacks and are inefficient for the precise detection of leukemia. This research proposes a deep learning-based framework for precise and automatic leukemia identification using microscopic images. The proposed framework comprises four stages which are pre-processing, data augmentation, segmentation, and the classification of leukemia. Moreover, pre-processing is utilized to clean the dataset images and eliminate the noise. Following that, data augmentation approaches have been employed to increase the number of images, and remove the class imbalance, and overfitting problems. The modified Convolutional Neural Network (CNN) based model is employed to segment the leukemia images. A well-known pre-trained AlexNet architecture has been used for classification. Besides that, a publicly available dataset Acute Lymphoblastic Leukemia Image DataBase (ALL-IDB) has been utilized to train and test the proposed model. The proposed model yielded 98.05% accuracy, a specificity of 97.59%, 100% of recall, and a 99.06% of F1-score. The experimentation results demonstrate that this model is effective and reliable for leukemia identification using the ALL-IDB dataset and suitable for deployment in clinical applications.

Keywords—Leukemia, CNN, Segmentation, ALL-IDB, Deep Learning

I. INTRODUCTION

Artificial Intelligence innovations are rapidly impacting our daily lives. Besides that, numerous contributions in the field of healthcare and medical science are being made including artificial intelligence advancements for the diagnosis and instantaneous treatment of diseases [1-3].

The identification and detection of diseases including hypertensive retinopathy [4, 5], Papilledema [6, 7], Brain Tumor [8-10], Glaucoma [11-13], Melanoma [14, 15], Alzheimer's [16], Central Serous Retinopathy [17, 18], Leukemia [19], and Liver Cancer [20] might be performed through Machine Learning (ML) and Deep Learning (DL).

Blood is considered a prime component of the human body. There are three kinds of blood cells in the human body named: White Blood Cells (WBCs), Red Blood Cells (RBCs), and Platelets. Moreover, these cells provide protective roles against various infections in the human body.

Similarly, the human body creates billions of new blood cells every day, the majority of which are red blood cells.

Leukemia is a cancer of blood cells that is accompanied due to unusual (irregular) growth of uncontrollable WBCs. However, the core reason for leukemia occurrence is the abundance of blast cell production in the bone marrow. Moreover, this rapid growth of WBCs damages the body's immune system [21]. Leukemia is one of the harmful cancers that is considered to be very threatening if not treated accurately. As per a report by the World Health Organization, around 900,000 cases are reported of blood cancer annually around the globe [22]. However, the majority of the impacted individuals are oblivious to this cancer as symptoms do not reveal at an early stage. Fig. 1, depicts healthy and blasts cell images.

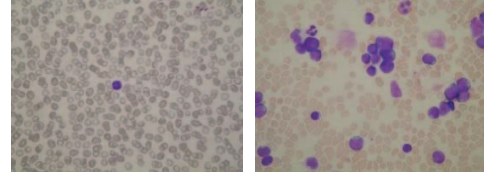


Fig. 1. Blood cells (a) healthy cells (b) leukemia affected

According to hematologists, leukemia is categorized into two classes: 1) acute leukemia and 2) chronic leukemia. Acute leukemia develops more quickly compared to chronic leukemia. Depending upon the cell type, acute and chronic leukemia are additionally categorized into four primary types namely, 1) Acute Myeloid Leukemia (AML), 2) Acute Lymphocytic Leukemia (ALL), 3) Chronic Myeloid leukemia (CML), and 4) Chronic Lymphocytic Leukemia (CLL) [22].

Leukemia identification at an early stage is crucial for the patient's recovery. Traditionally, the detection of leukemia was conducted manually by hematologists and pathologists by perceiving the blood smear images. However, this manual detection requires trained and experienced pathologists to identify the cancerous cells. Typically, manual approaches are difficult, time-consuming, and produce inaccurate results [23]. Nevertheless, leukemia can now be detected and treated using Artificial Intelligence approaches. Additionally, the manual leukemia examination has been replaced by ML and DL-based automated systems that may reduce human errors and save time. Furthermore, there exist some cutting-edge technologies to detect leukemia

including [24-28]. Consequently, there are still some research gaps to design an automated and reliable system to detect the presence of leukemia. This study marks an important development toward the detection of leukemia through microscopic images. A CNN-based framework has been proposed in this study to diagnose leukemia using microscopic images. Initially, to eliminate the noise from the dataset pre-processing has been adapted, and to increase the dataset size, various data transformation approaches have been utilized. Apart from that, the proposed CNN model segment the images and classifies them as normal or blast cells. The major contributions of this research are stated below.

A modified CNN model is proposed in this research for leukemia segmentation. The proposed framework contains various CNN layers and is trained on two datasets to enhance the model's performance. Moreover, the images of the dataset are pre-processed to remove the noise and for data cleaning. Additionally, different data transformation techniques have been used to address the small dataset and class imbalance issue. Furthermore, the existing literature has computational complexity constraints that have been addressed in the proposed approach. Finally, a pre-trained AlexNet model was applied to the segmented images for cell classification and achieved efficient outcomes.

The remaining section is organized as follows: section II elaborates a comprehensive review of the literature. A proposed method for leukemia identification through microscopic has been covered in section III. The experimental outcomes and the comparison of the proposed model is elaborated in section IV. Lastly, the conclusion of the research is described in section V.

II. LITERATURE REVIEW

The advancement in artificial intelligence technologies and various ML and DL algorithms for leukemia identification have been mentioned in the existing literature. Abdeldaim et al. [29] developed a comprehensive framework to recognize the ALL from blood images utilizing the ALL-IDB2 dataset comprised of 260 images. A machine learning K-NN algorithm was employed for the classification process that achieved 96.01% classification accuracy. Furthermore, a CNN-based technique having seven crucial layers was proposed in [30]. In the proposed methodology, primarily five layers were employed for feature extraction and the rest of the layers were for classification. A publicly available ALL-IDB1 dataset with 108 images was used to train the

model which yielded 96.6% accuracy. Another CNN architecture named ConVNet was proposed in [31] and achieved 81.74% accuracy. Additionally, the proposed multilayer architecture was utilized to extricate the features. Moreover, the proposed CNN framework outperformed than Support Vector Machine (SVM), Multilayer Perceptron (MLP), and Random Forest (RF). Al-jaboriy et al. [32] designed a method based on ML to segment the ALL from blood stain images. ALL-IDB1 dataset was utilized for training the model, however, Artificial Neural Network (ANN) was used for the classification. The suggested algorithm attained 97.07% accuracy. In addition, a hybrid framework proposed by Jha et al. in [33] utilizes Mutual Information (MI) for segmenting leukemia-affected cells. A Chronological-Sine, Cosine-Algorithm (SCA) was used for classification purposes and achieved an accuracy of 98.7%. Das et al. in [34] proposed an automatic leukemia classification framework using a Gini index-based Fuzzy Naive Bayes (GFNB) classifier. ALL-IDB1 dataset was employed for training the proposed model and yielded 95% accuracy. An advanced hybrid technique was introduced by sahlol et al. in [35] to diagnose ALL. A CNN architecture named VGG-16 was employed for feature extraction from WBCs. Following that, Statistically Enhanced Salp Swarm Algorithm (SESSA) was used to filter the extricated features. ALL-IDB2 and C-NMC-2019 datasets were used to train the model and achieved 96.11% and 87.9% accuracy respectively. A You Only Look Once (YOLOv4), the object detection algorithm was proposed by Khandekar et al. in [36] for leukemia detection. The proposed framework achieved 96.06% and 98.7% classification accuracy for ALL-IDB1 and C-NMC-2019 datasets respectively. In addition, a robust architecture was also designed in [37] for ALL detection. Likewise, many CNN models were compared, with VGG-16 achieving the highest accuracy of 96.84%. A CAD-3 framework proposed by Abas et al. [38] consists of three Machine Learning models. However, You Only Look Once (YOLOv2) was utilized for the WBCs detection, while a CNN architecture was employed for selected cell classification. ALL-IDB1 and BCCD dataset was employed which achieved 94.3% leucocytes classification accuracy. Moreover, a CNN-based architecture for leukemia identification was proposed in [39]. The proposed model begins with data augmentation and squeeze and excitation learning was utilized to enhance the features. Two freely available datasets named ALL-IDB1 and ALL-IDB2 were utilized for experimentation and yielded 98.3% overall accuracy. A summary of existing literature to diagnose leukemia is shown in Table I.

TABLE I. SUMMARY OF LITERATURE REVIEW

Ref. no.	Year	Authors	Datasets	Results
[29]	2018	Abdeldaim et al.	ALL-IDB2	96.01% accuracy
[30]	2018	Thanh et al.	ALL-IDB1	96.6% accuracy
[31]	2019	Pansombut et al.	ALL-IDB1 & ASH image bank	81.74% average accuracy
[32]	2019	Al-jaboriy et al.	ALL-IDB1	97.07% accuracy
[33]	2019	Jha et al.	ALL-IDB2	98.7% accuracy
[34]	2020	Das et al.	ALL-IDB1	Specificity of 1%, accuracy of 95%, and sensitivity of 0.95%
[35]	2020	Sahlol et al.	ALL-IDB2 and C-NMC-2019	Achieved 96.11% and 87.9% classification accuracy respectively
[36]	2021	Khandekar et al.	ALL-IDB1 and C-NMC-2019	Accuracy of 96.06% and 98.7%, recall of 92% and 96% respectively

[37]	2021	Genovese et al.	ALL-IDB2	Accuracy of 96.84%, 3.15 error rate, the sensitivity of 97.53%, and specificity of 96.15%
[38]	2022	Abas et al.	ALL-IDB1 and BCCD	96% average precision and 94.3% average accuracy
[39]	2022	Bukhari et al.	ALL-IDB1 and ALL-IDB2	98.3% of accuracy, 98.16% of precision, 98.16% of recall, and 98.43% of F1-score

III. PROPOSED METHODOLOGY

A deep CNN-based framework has been proposed for leukemia detection employing the ALL-IDB dataset. The proposed model consists of four basic steps; pre-processing,

data augmentation, segmentation, and classification. The proposed model takes the microscopic image as an input and classifies whether the image has normal cells or blast cells. Fig. 2, depicts the flow of the proposed methodology.

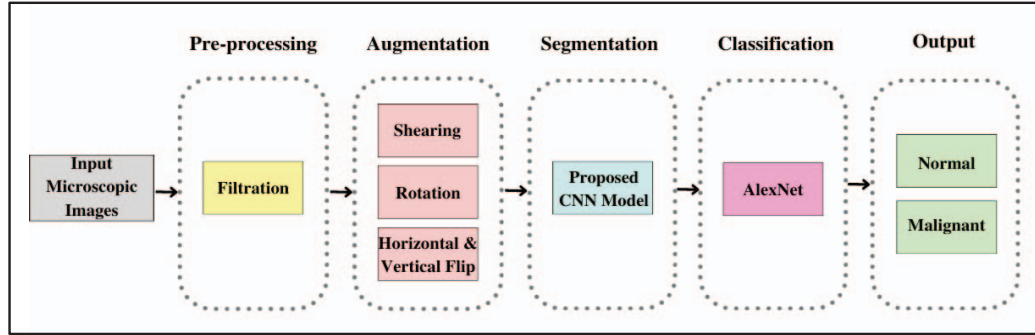


Fig. 2. Workflow diagram of the proposed methodology for leukemia classification

A. Dataset

In this study, a publicly available dataset named: Acute Lymphoblastic Leukemia Image Database (ALL-IDB) has been employed to train and test the proposed model [40]. ALL-IDB dataset is divided into two categories: 1) ALL-IDB1 and 2) ALL-IDB2. Whereas, both of these datasets are focused on image segmentation and classification. ALL-IDB1 dataset is made up of 108 original blood cell RGB images, of which 59 images are normal and the remaining 49 are leukemia-affected. All the images have a 24-bit color depth and are in JPG format. Moreover, these images were captured in September 2005. It holds around 39000 blood components, in which lymphocytes have been identified by oncologists. Subsequently, the ALL-IDB2 is a dataset of a cropped region of interest that contains healthy and malignant images and it relates to ALL-IDB1. This dataset comprises of total 260 images, however, 50% of these images are lymphoblasts and the remaining 50% are normal images. Additionally, all the images are in JPG format and it holds 257 * 257 resolution.

B. Pre-processing

Data pre-processing is a common step in the DL model to improve image quality. Its basic purpose is to magnify the quality of images, eliminate the data's noise, and normalize the contrast of microscopic images. The noise is eliminated from microscopic images to allow the system to get useful data. However, a filtration method has been used to eliminate the background impurities.

1) *Filtration*: Filtration is a technique to modify and remove the background contamination from microscopic images. Therefore, the median filter has been used for noise removal purposes and to sharpen the images. It is a non-linear filtering operation having a 3 * 3 kernel size to attain sharp

features from microscopic images. The normal and filtered images have been depicted in Fig. 3.

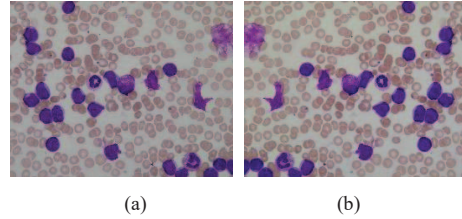


Fig. 3. Normal and pre-processed leukemia images

C. Augmentation

Data transformation is used to increase the dataset's images. Additionally, it also addresses the problem of overfitting, which happens because of the limited number of training images [41]. Various data transformation methods such as shearing, rotation, and horizontal and vertical flip were applied to attain several variants of the pre-processed images.

1) *Shearing*: Shearing alters the orientation of the original image. This approach shifts the pixels of the image in a clockwise direction by using specified angles. This study utilized a 45-degree shearing angle for image transformation.

2) *Rotation*: An image is rotated between 0 to 360 degrees in a clockwise direction in the rotation technique. The 120-degree rotation was used in this study for image transformation.

3) *Horizontal and vertical flip*: In this method of flipping, the pixels move either horizontally or vertically.

The original datasets ALL-IDB1 and ALL-IDB2 contain 108 and 260 images respectively, therefore, the data augmentation technique was much needed to train the deep

$$\text{Specificity} = \frac{TN}{TN+FP} \quad (2)$$

$$\text{Recall or sensitivity} = \frac{TP}{TP+FN} \quad (3)$$

$$\text{F1-score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (4)$$

TABLE II. COMPARISON OF PROPOSED MODEL WITH ALL-IDB DATASET

Ref. no.	Techniques	Datasets	Results
[29]	Zack algorithm	ALL-IDB2	96.01% accuracy
[32]	Genetic Algorithm (GA) and ANN	ALL-IDB1	97.07% of classification accuracy
[34]	Gini index-based Fuzzy Naive Bayes (GFNB)	ALL-IDB1	Accuracy of 95%
[37]	CNN architecture (VGG16)	ALL-IDB2	96.84% of accuracy, a sensitivity of 97.53%, a specificity of 96.15%, and a 3.15 error rate
[38]	CAD3 framework	ALL-IDB1 and BCCD	96% average precision and 94.3% of average accuracy
Proposed method for leukemia segmentation and classification	CNN for segmentation, and AlexNet for classification	ALL-IDB	Accuracy of 98.05%, specificity of 97.59%, recall of 100%, and 99.06% of F1-score

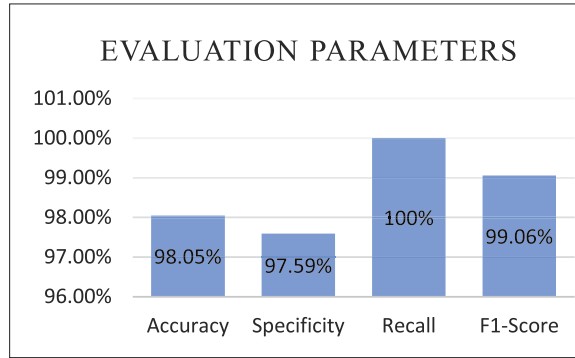


Fig. 6. Graphical representation of results

V. CONCLUSION

A deep learning-based automatic leukemia detection framework has been proposed in this research. Moreover, a binary classification approach has been utilized to identify normal and malignant cells from blood smear images. Initially, images were filtered, and eliminated the background noise to clean the microscopic data in pre-processing phase. Additionally, several data transformation techniques have been employed to overcome the small dataset and class imbalance issues. A publicly available ALL-IDB dataset was employed for the model's training and testing. In addition, a modified CNN-based framework has been proposed that consists of different layers such as convolutional layers, batch normalization, leaky ReLU (activation function), and max-

pooling layers to segment the leukemia images. Finally, a CNN model AlexNet was utilized to classify leukemia from microscopic images. The proposed framework efficiently classified the images as normal or leukemia-affected images. The experimental outcomes depict that the proposed framework attained an accuracy of 98.05%, specificity of 97.59%, 100% of recall, and 99.06% of F1-score. The proposed automated approach can assist pathologists in detecting leukemia at an early stage and saving patients' lives. In the future, the proposed framework may be extended to segment the sub-types of leukemia from blood microscopic images.

CONFLICT OF INTEREST

The author of this research has no conflict of interest.

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